

Problem – Solution Fit Canvas

Project Design Phase Report

Date	15 February 2025	Maximum Marks	2 Marks
Team ID	TEAM-SN-2026	Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value

Overview & Purpose

Definition: The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer’s problem. It helps entrepreneurs, administrators, and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and security strategy with the right triggers and messaging.
- Increase touch-points with your organization by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly security vulnerabilities.
- Understand the existing operational situation in order to improve it for your target user groups.

Problem – Solution Fit Details

1. CUSTOMER SEGMENT(S) [CS] • Primary Target: ServiceNow System Administrators & Security Engineers responsible for access control policies. • Secondary Target: IT Service Managers, Process Owners, and Governance & Compliance Officers. • Context: Enterprise organizations running ServiceNow instances requiring dynamic, field-level data protection.	2. JOBS-TO-BE-DONE / PROBLEMS [J&P] • Job-to-be-Done: Dynamically restrict record read, write, and execute permissions based on runtime field values (e.g., locking restricted fields when a ticket reaches 'Closed' or 'In Progress' state). • Problem: Out-of-the-box (OOTB) static role-based ACLs cannot evaluate complex, multi-condition script logic at runtime across all data interfaces.
3. TRIGGERS [TR] • Internal Triggers: Internal security audit findings flagging unauthorized record visibility or field tampering in inactive ticket states. • External Triggers: Compliance updates (OWASP / ISO 27001) requiring strict server-side access controls on sensitive IT service records.	4. EMOTIONS: BEFORE / AFTER [EM] • Before Solution: Anxious about security breaches, frustrated by rigid static security roles, and worried about failing data governance audits. • After Solution: Confident in platform data integrity, assured during compliance reviews, and satisfied with automated, dynamic access control.
5. AVAILABLE SOLUTIONS (ALTERNATIVES) [AS] • Alternative 1 - UI Policies / Client Scripts: Hides or disables fields visually on forms, but fails to prevent	6. CUSTOMER CONSTRAINTS [CC] • Technical Constraints: Strict database query performance budgets; unoptimized ACL script logic can degrade list loading and report generation times.

backend unauthorized access via REST/SOAP APIs or Reporting engines. • Alternative 2 - Static Role ACLs: Creates 'role explosion' by requiring separate custom security roles for every possible combination of field states.	• Privilege Constraints: Requires elevated security privileges (security_admin role) to configure and deploy ACL rules in production.
7. BEHAVIOUR [BE] • Direct Action: Admins attempt to rely solely on client-side form rules, unintentionally leaving REST API endpoints exposed to unauthorized modifications. • Indirect Action: Security teams perform manual, labor-intensive log audits after unauthorized field changes occur.	8. CHANNELS OF BEHAVIOUR [CH] • 8.1 Online Channels: ServiceNow Developer Portal, Product Documentation, ServiceNow Community Forums, and Instance Admin Workspaces. • 8.2 Offline Channels: IT Governance Boards, Architecture Advisory Boards (CAB), Security Review Meetings, and Team Sprint Demos.
9. PROBLEM ROOT CAUSE [RC] • Root Cause Analysis: Standard declarative ACL Condition Builders lack dynamic evaluation capabilities to check runtime field values against user session properties without executing server-side JavaScript logic.	10. YOUR SOLUTION [SL] • Core Solution: Script-Controlled Access Control List (ACL) Rule (sys_security_acl) utilizing server-side JavaScript execution. • Key Capability 1: Evaluates runtime field states (e.g., current.state, current.assignment_group) against user attributes (gs.getUser()) directly on the server. • Key Capability 2: Enforces universal security across all data entry points: Web UI, Next Experience, REST APIs, and Reporting Engines. • Optimization: Lightweight script evaluation designed without heavy GlideRecord loops to maintain peak platform performance.

References:

1. Problem-Solution Fit Canvas Framework: <https://www.idealhackers.network/problem-solution-fit-canvas/>
2. Problem-Solution Fit Canvas Guide: <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>
3. ServiceNow Developer Portal: <https://developer.servicenow.com/>